

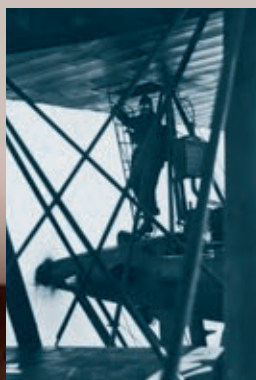


“GIANT AIRCRAFT”

The massive scale of the “R-planes” is shown by this image of an R.IV surrounded by people. Its wingspan measured 138ft (42m) and it weighed over 28,600lb (13,000kg).

DAREDEVIL MECHANICS

If the aircraft’s engines needed attention, mechanics would move around its airframe even in midflight.



calculations could be thrown completely by the effect of wind. If navigators failed to realize that they had been blown off course, they were lost and had no way to regain their bearings.

Of course, a fair number of airships heading for London did find it. The city’s lights were hidden by a strictly enforced blackout, but they were able to use the Thames River as a navigational aid, since it offered a readily identifiable shape for them to follow. They also had searchlight and gun batteries marked on their maps, not only so they could avoid them but also so they could use them for orientation around the darkened city.

Airships did not attack a target in formation. Each came from a different direction, at staggered intervals. Once through the ring of defenses around London, the method of attack was simple – to fly as fast as possible in a straight line across the city, releasing their explosive and incendiary bombs as they went.

The first airship raids on Britain were carried out in January 1915; the first raid on London the following May. Up to the late summer of 1916, although sporadic, the raids could be counted a success. Some inflicted a substantial amount of damage – for example, on September 8, 1915, Captain Heinrich Mathy’s L 13 killed 22 people and injured 87 in a single pass across London. Even wandering lost over England in the middle of the night was not necessarily a waste of energy. One of the worst raids of the war came in January 1916, when nine airships trying to bomb Liverpool became confused in the darkness and arrived instead over the cities of the Midlands, where no blackout was in force. 70 people were killed in the bombing.

The British were forced to divert valuable resources of aircraft, pilots, and guns from the

Western Front as the public demanded better protection. But these resources, accompanied by the crucial introduction of explosive and incendiary ammunition, succeeded in tipping the scales fatally against the Zeppelins. In late August and September 1916, the Germans launched their most ambitious airship bombing offensive. It was a disaster. In a series of raids, four airships were lost to ground fire or pursuit aircraft. Among those who died was the experienced Mathy; he jumped to his death from his burning Zeppelin rather than roast alive.

High fliers

The Germans responded to this setback with a new type of airship that was bigger but lighter than previous ones. These “height-climbers” regularly flew at over 16,000ft (5,600m), out of

reach of ground fire and difficult for aircraft to attack. But for the unfortunate crews, missions at high altitude were a severe trial. For hours on end they endured freezing cold; they found it hard to breathe in the thin air but using their crude oxygen equipment made them feel nauseated; rapid changes of pressure in ascending and

“Inside the fuselage the pale glow of dimmed lights outlines the chart table, the wireless equipment, and the instrument panel... Under us is a black abyss.”

HAUPTMANN ARTHUR SCHOELLER

COMMANDER OF AN R-PLANE, DESCRIBING A NIGHT FLIGHT TO BRITAIN IN MARCH 1918

descending gave them the bends. And for all this suffering they inflicted only limited damage.

However, the technical achievements of German airships were extraordinary. In November 1917, the L 59 flew 4,200 miles (6,760km) nonstop on an abortive mission to Africa. In the same year, L 55 established an enduring altitude record of 24,000ft (8,400m).

But turning technical capability into military effectiveness proved impossible. To the end, naval airship leader Peter Strasser still dreamed of a decisive coup – an airship raid across the Atlantic to devastate New York. However, in August 1918 he was shot down on a final hopeless mission over the North Sea.

SKY JUMPER

In this stereoscopic photograph, British soldiers look at the hole left in the ground by the body of a man who jumped from a burning Zeppelin over Billericay, Essex, England. Using a stereoscopic viewer, it is possible to see the hole in 3-D.

